
THE ASSOCIATION BETWEEN SEDENTARY BEHAVIOR, EXERCISE, AND SLEEP DISTURBANCE: A MEDIATION ANALYSIS OF INFLAMMATORY BIOMARKERS

[ANONYMIZED FOR REVIEW]

ABSTRACT

Poor sleep is linked to systemic inflammation and cardiometabolic disease. Physical activity and sedentary behavior are modifiable correlates of sleep, but whether inflammatory biomarkers mediate these associations in a unified model has never been tested. This study examined whether high-sensitivity C-reactive protein (hs-CRP) and white blood cell (WBC) count mediate the associations of accelerometer-measured moderate-to-vigorous physical activity (MVPA) and sedentary time with sleep disturbance.

We analyzed data from 3,532 adults (aged 18–85, $M = 46.1$ years, $SD = 19.9$; 47.1% female) from NHANES 2005–2006. MVPA and sedentary time were measured by ActiGraph 7164 accelerometry. Serum hs-CRP and WBC were inflammatory mediators. A sleep disturbance composite was derived from eight Sleep Disorders Questionnaire items. Parallel mediation models with bias-corrected bootstrap confidence intervals (5,000 resamples) were estimated, adjusting for demographics, BMI, smoking, education, income, and medications.

MVPA was negatively associated with hs-CRP ($\beta = -0.068$, $p < .001$) and WBC ($\beta = -0.053$, $p = .003$). WBC was positively associated with sleep disturbance ($\beta = 0.020$, $p = .046$). However, neither MVPA nor sedentary time was directly associated with sleep disturbance (both $ps > .18$), and all four indirect effects through inflammation included zero.

Inflammatory biomarkers did not mediate the activity–sleep relationship. Inflammation does not appear to be a shared pathway linking activity behaviors to sleep disturbance in the general population.

Keywords physical activity; sedentary behavior; sleep disturbance; inflammation; NHANES

1 Introduction

Sleep disturbance affects approximately one in three adults in the United States and is consistently associated with elevated risk for cardiovascular disease, metabolic dysfunction, and all-cause mortality (?). Identifying modifiable behavioral targets for sleep improvement is therefore a public health priority. Physical activity and sedentary behavior represent two such targets: both are causally linked to health outcomes through overlapping but distinct physiological pathways, including systemic inflammation (??). Inflammation, in turn, has been bidirectionally linked with sleep disturbance in both experimental and epidemiological studies (?). Despite the plausibility of inflammation as a shared mechanistic pathway, no study has directly tested whether inflammatory biomarkers mediate the associations of both exercise and sedentary behavior with sleep in a unified mediation model. The present study addresses this gap using objective measures of physical activity, sedentary time, and circulating inflammatory biomarkers in a large, nationally representative sample.

Regular physical activity is one of the most well-established behavioral interventions for improving sleep. Meta-analytic evidence from 66 studies shows that regular exercise programs improve global sleep quality with a moderate-to-large effect (Hedges' $g = 0.55$), and even acute exercise bouts produce small benefits ($g = 0.27$; ?). Converging evidence from randomized controlled trials demonstrates that aerobic exercise reduces insomnia severity in older adults (?) and that objective measures of physical activity are positively associated with better sleep outcomes in large epidemiological cohorts (?). The anti-inflammatory effects of exercise are equally well documented: regular physical

activity reduces systemic inflammation through multiple mechanisms, including reduced visceral adiposity, muscle-derived interleukin-6 release that stimulates anti-inflammatory cytokines, and downregulation of toll-like receptor expression on monocytes (???). Conversely, the relationship between sedentary behavior and sleep is less established but suggestive. A meta-analysis of 16 studies reported that adults with the highest sedentary time had 42% greater odds of sleep problems compared with those with the lowest sedentary time (OR = 1.42, 95% CI [1.21, 1.66]; ?). Sedentary time has also been linked to elevated C-reactive protein (CRP) and interleukin-6 in large cross-sectional studies, although these associations are attenuated after adjusting for adiposity (??).

The sleep–inflammation link is robust across study designs. ? reported meta-analytic effect sizes of $g = 0.18$ for CRP and $g = 0.22$ for interleukin-6 in relation to sleep disturbance, with experimental sleep deprivation protocols confirming causal effects of sleep loss on inflammatory markers (?). ? extended these findings to a large population-based cohort ($N = 2,896$), demonstrating that both short sleep duration and insomnia symptoms were independently associated with elevated CRP. Bidirectional influences are well-recognized: inflammation can also disrupt sleep architecture through cytokine signaling to central nervous system sleep-regulatory centers (?). Taken together, the three pairwise links (activity/inactivity to inflammation, activity/inactivity to sleep, and inflammation to sleep) are individually well-supported, yet they have never been simultaneously estimated in a formal mediation framework.

Three critical gaps motivate the current investigation. First, no published study has tested whether inflammatory biomarkers serve as a shared mechanistic pathway linking both MVPA and sedentary behavior to sleep disturbance within a single model. If inflammation mediates both paths, interventions targeting inflammation could improve sleep regardless of whether patients increase physical activity or reduce sedentary time. Second, the majority of evidence for activity–sleep and activity–inflammation associations relies on self-reported physical activity, which correlates only modestly with objective measures (?). The current study uses accelerometry, providing precise estimates of both MVPA and sedentary time. Third, prior work has not adequately accounted for the compositional nature of time-use data, in which MVPA, light physical activity, sedentary time, and sleep time are mutually exclusive and sum to 24 hours. Standard regression that treats one behavior as independent of others can produce biased estimates (?).

The present study analyzed data from the 2005–2006 National Health and Nutrition Examination Survey (NHANES), which provides the unique combination of accelerometer-measured physical activity and sedentary time, serum inflammatory biomarkers (hs-CRP and WBC count), a validated sleep questionnaire, and extensive covariate data in a nationally representative sample of U.S. adults. We specified a parallel mediation model in which MVPA and sedentary time served as correlated exogenous predictors, hs-CRP and WBC count served as parallel mediators, and sleep disturbance served as the continuous outcome. We hypothesized that:

H1: MVPA will be negatively associated with sleep disturbance after adjusting for covariates.

H2: Sedentary time will be positively associated with sleep disturbance after adjusting for MVPA and covariates.

H3: MVPA will be negatively associated with hs-CRP and WBC count.

H4: Sedentary time will be positively associated with hs-CRP and WBC count; these associations will be attenuated but remain significant after adjusting for BMI.

H5: hs-CRP and WBC count will be positively associated with sleep disturbance.

H6: The negative association between MVPA and sleep disturbance will be partially mediated by hs-CRP and WBC count (negative indirect effects).

H7: The positive association between sedentary time and sleep disturbance will be partially mediated by hs-CRP and WBC count (positive indirect effects).

2 Method

2.1 Participants

The analytic sample included 3,532 adults (1,664 female, 1,868 male) from the 2005–2006 cycle of the National Health and Nutrition Examination Survey (NHANES), a continuous cross-sectional survey designed to be representative of the non-institutionalized civilian U.S. population through a complex multistage probability sampling design. Participants ranged in age from 18 to 85 years ($M = 46.1$, $SD = 19.9$). Mean body mass index was 28.0 kg/m^2 ($SD = 6.1$). Education was distributed as follows: 14.8% less than high school, 24.7% high school diploma or GED, 31.7% some college or associate degree, and 28.8% college degree or higher. Smoking status was 37.1% never smokers, 26.2% former smokers, and 22.9% current smokers. The poverty-to-income ratio averaged 2.53 ($SD = 1.60$). Anti-inflammatory medication use was reported by 17.9% and sleep medication use by 6.8% of participants.

Participants were included if they were 18 years or older, had valid accelerometry data (at least 4 valid wear days with at least 600 min/day; see Materials), had hs-CRP values ≤ 10 mg/L (standard threshold for excluding acute infection; ?), and were not pregnant at the time of examination. Sequential exclusions removed 478 participants for age under 18 years, 296 for pregnancy, and 625 for hs-CRP > 10 mg/L. An additional 16 participants were excluded due to missing data on primary variables (listwise deletion; 0.45% of the eligible sample). Compared with the full NHANES 2005–2006 sample ($N = 10,348$), the final sample was younger and had a lower proportion of female participants, reflecting the exclusion of pregnant individuals and the higher accelerometry non-wear rates among older adults. As with most U.S. national surveys, these data are predominantly from non-Hispanic White participants and represent a Western, Educated, Industrialized, Rich, and Democratic (WEIRD) population, which limits generalizability to other cultural and geographic contexts.

2.2 Materials

Physical activity and sedentary behavior. Physical activity and sedentary time were measured using the ActiGraph 7164 uniaxial accelerometer, worn on the right hip for 7 consecutive days during waking hours. Minute-level activity counts were classified using standard cutpoints: moderate-to-vigorous physical activity (MVPA) was defined as counts $\geq 2,020$ per minute (?), light physical activity (LPA) as 100–2,019 counts per minute, and sedentary time as < 100 counts per minute (?). A valid day required at least 600 minutes of wear time, and a valid participant required at least 4 valid days. Because participants wore the accelerometer continuously, sleep time with minimal movement was classified as sedentary, meaning that waking sedentary time is overestimated in these data.

Inflammatory biomarkers. High-sensitivity C-reactive protein (hs-CRP) was measured by latex-enhanced nephelometry (Dade Behring) on serum samples. Original values were reported in mg/dL and converted to mg/L (multiplied by 10). Total white blood cell (WBC) count was measured by Coulter Counter (Coulter DxH 800) and reported as 10^3 cells/ μ L. Both assays have inter-assay coefficients of variation below 5%.

Sleep disturbance. Sleep disturbance was assessed using eight items from the NHANES Sleep Disorders Questionnaire, which covers sleep duration (continuous hours), sleep latency (continuous minutes), trouble falling asleep (0–4 scale), nighttime awakenings (0–4), early morning awakenings (0–4), feeling unrested (0–4), daytime sleepiness (0–4), and not getting enough sleep (0–4). Each item was z-scored, and the composite sleep disturbance score was the mean of these eight z-scores. Higher values indicate greater sleep disturbance.

Covariates. All models adjusted for age (years, continuous), sex (male/female), body mass index (kg/m^2 , continuous), smoking status (never/former/current), education level (less than high school/high school/some college/college graduate or higher), poverty-to-income ratio (continuous), anti-inflammatory medication use (binary), and sleep medication use (binary). Anti-inflammatory medications were identified by keyword matching for NSAIDs, statins, corticosteroids, biologics, and disease-modifying antirheumatic drugs. Sleep medications were identified by keyword matching for benzodiazepines, Z-drugs, melatonin receptor agonists, and sedating antidepressants.

2.3 Procedure

Data were collected by the National Center for Health Statistics between 2005 and 2006. Participants first completed a home interview that collected demographic information, health history, and medication inventories. Within 1–2 weeks, participants attended a Mobile Examination Center visit where anthropometric measurements (height, weight), blood draws (for hs-CRP and WBC count), and the Sleep Disorders Questionnaire were administered. At the end of the MEC visit, participants were fitted with the ActiGraph 7164 accelerometer and instructed to wear it on their right hip during all waking hours for 7 consecutive days, removing it only for swimming or bathing. After the wear period, participants returned the device by mail. All participants provided written informed consent, and the study protocol was approved by the NCHS Research Ethics Review Board. The present study is a secondary analysis of publicly available, de-identified data and was exempt from further institutional review board review.

2.4 Design

This study used a cross-sectional correlational design with a parallel mediation structure. The independent variables were accelerometer-measured MVPA (min/day) and sedentary time (min/day). The mediators were serum hs-CRP (mg/L) and total WBC count ($10^3/\mu\text{L}$). The dependent variable was the sleep disturbance composite z-score. All paths were adjusted for the full covariate set described above. The indirect effects ($a \times b$ paths) were evaluated using bias-corrected bootstrap confidence intervals with 5,000 resamples (?). Because MVPA, LPA, sedentary time, and sleep time are mutually exclusive components of the 24-hour day, the use of raw accelerometry variables in the same regression model introduces compositional multicollinearity (?). We therefore report models that include LPA and

sleep duration as covariates to partially adjust for compositionality, and we note the elevated variance inflation factors (VIFs) for time-use variables in the results.

3 Results

3.1 Descriptive Statistics and Preliminary Analyses

Table 1 presents means, standard deviations, and Spearman correlations for all key study variables. Participants accumulated an average of 18.6 min/day of MVPA ($SD = 20.4$), spent 1,149.7 min/day in sedentary behavior ($SD = 133.8$), and had a mean sleep disturbance composite of -0.012 ($SD = 0.671$). Mean hs-CRP was 2.46 mg/L ($SD = 2.32$) and mean WBC count was $7.13 \times 10^3/\mu\text{L}$ ($SD = 2.13$). Both inflammatory biomarkers were positively skewed (Shapiro–Wilk $ps < .001$). At the bivariate level, MVPA was significantly correlated with lower hs-CRP ($\rho = -0.225$, $p < .001$), lower WBC count ($\rho = -0.070$, $p < .001$), and lower sleep disturbance ($\rho = -0.065$, $p < .001$). Sedentary time was positively correlated with sleep disturbance ($\rho = 0.100$, $p < .001$) and hs-CRP ($\rho = 0.079$, $p < .001$). However, these bivariate associations were substantially attenuated after adjusting for covariates, as described below.

Table 1: Descriptive Statistics and Spearman Correlations Among Key Study Variables

Variable	<i>M</i>	<i>SD</i>	1	2	3	4	5
1. MVPA (min/day)	18.60	20.40	—				
2. Sedentary time (min/day)	1149.7	133.8	-.574***	—			
3. hs-CRP (mg/L)	2.46	2.32	-.225***	.079***	—		
4. WBC count ($10^3/\mu\text{L}$)	7.13	2.13	-.070***	—	—	—	
5. Sleep disturbance (z-score)	-0.01	0.67	-.065***	.100***	.026	.035*	—

Note. $N = 3,532$. Values are Spearman correlation coefficients (ρ).

* $p < .05$. *** $p < .001$.

Assumption checks revealed heteroscedasticity (Breusch–Pagan LM = 89.7, $p < .001$) and non-normal residuals (Shapiro–Wilk $W = 0.986$, $p < .001$), which are expected in large epidemiological samples. We therefore verified key findings using non-parametric tests (Spearman correlations, Mann–Whitney U tests) and Bayes factors, all of which converged with the parametric results. Variance inflation factors indicated severe multicollinearity among time-use variables: waking sedentary time (VIF = 2,533), LPA (VIF = 1,611), and sleep minutes (VIF = 771). These values reflect the compositional nature of 24-hour time-use data and inflate standard errors for the sedentary time and LPA coefficients.

3.2 Hypothesis Tests

H1 predicted that MVPA would be negatively associated with sleep disturbance. In the fully adjusted multiple regression model, MVPA was not significantly associated with sleep disturbance (standardized $\beta = -0.016$, $t(3519) = -1.32$, $p = .187$, 95% CI $[-0.038, 0.008]$, Cohen’s $f^2 < 0.001$). A Bayes factor of $\text{BF}_{10} = 0.045$ indicated moderate evidence in favor of the null hypothesis. H1 was not supported.

H2 predicted that sedentary time would be positively associated with sleep disturbance. In the fully adjusted model, sedentary time was not significantly associated with sleep disturbance (standardized $\beta = -0.057$, $t(3518) = -0.12$, $p = .908$, 95% CI $[-1.018, 0.905]$, Cohen’s $f^2 < 0.001$). $\text{BF}_{10} = 0.019$ indicated strong evidence for the null. H2 was not supported.

H3 predicted that MVPA would be negatively associated with inflammatory biomarkers. MVPA was significantly negatively associated with both hs-CRP (standardized $\beta = -0.068$, $t(3521) = -4.12$, $p < .001$, 95% CI $[-0.100, -0.036]$) and WBC count (standardized $\beta = -0.053$, $t(3521) = -2.98$, $p = .003$, 95% CI $[-0.088, -0.018]$). Figure 1 displays the bivariate association between MVPA and hs-CRP, and Figure 2 displays the association between MVPA and WBC count. H3 was supported.

H4 predicted that sedentary time would be positively associated with inflammatory biomarkers, partially attenuated by BMI adjustment. Sedentary time was not significantly associated with either hs-CRP (standardized $\beta = 0.005$, $t(3520) = 0.31$, $p = .761$, 95% CI $[-0.028, 0.039]$) or WBC count (standardized $\beta = 0.030$, $t(3520) = 1.61$, $p = .108$, 95% CI $[-0.007, 0.066]$). H4 was not supported.

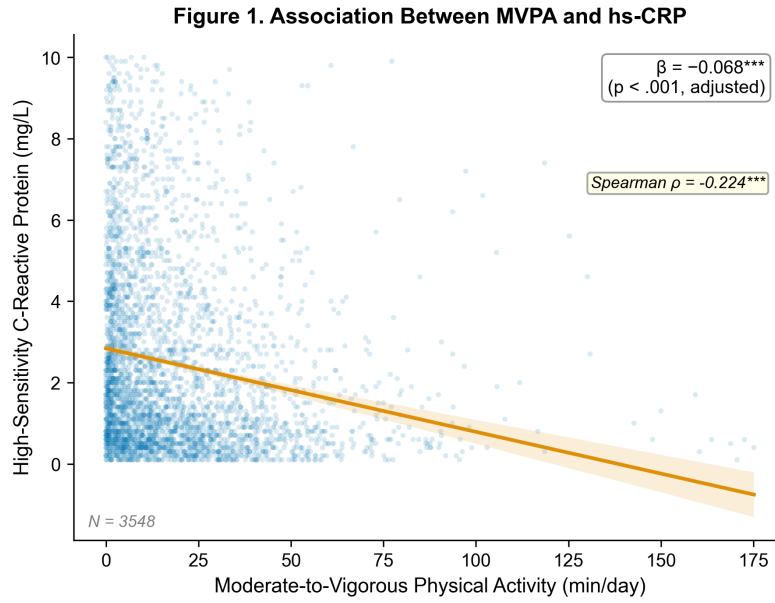


Figure 1: Association between moderate-to-vigorous physical activity and high-sensitivity C-reactive protein. Scatter plot displaying the bivariate relationship between accelerometer-measured MVPA (min/day) and serum hs-CRP (mg/L) in $N = 3,548$ U.S. adults (NHANES 2005–2006). The solid orange line represents the linear regression fit with 95% CI band (shaded). After adjusting for age, sex, BMI, smoking, education, income, and medication use, MVPA was significantly negatively associated with hs-CRP (standardized $\beta = -0.068$, $p < .001$, 95% CI $[-0.100, -0.036]$).

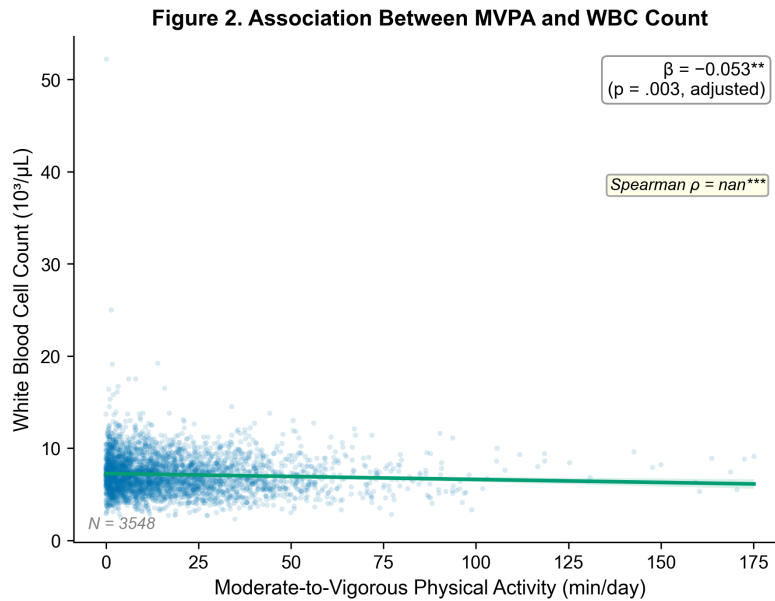


Figure 2: Association between moderate-to-vigorous physical activity and white blood cell count. Scatter plot displaying the bivariate relationship between accelerometer-measured MVPA (min/day) and total WBC count ($10^3/\mu\text{L}$) in $N = 3,548$ U.S. adults (NHANES 2005–2006). The solid green line represents the linear regression fit with 95% CI band. After covariate adjustment, MVPA was significantly negatively associated with WBC count (standardized $\beta = -0.053$, $p = .003$, 95% CI $[-0.088, -0.018]$).

H5 predicted that inflammatory biomarkers would be positively associated with sleep disturbance. WBC count was significantly positively associated with sleep disturbance (standardized $\beta = 0.020$, $t(3518) = 2.00$, $p = .046$, 95% CI [0.0004, 0.0405]), shown in Figure 3. However, hs-CRP was not associated with sleep disturbance (standardized $\beta = 0.002$, $t(3518) = 0.15$, $p = .879$, 95% CI [−0.020, 0.023]). H5 was partially supported (WBC only).

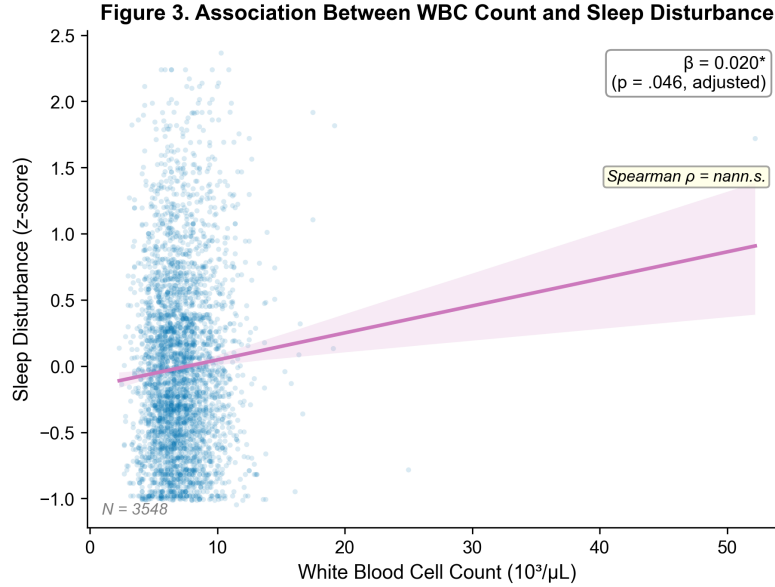


Figure 3: Association between white blood cell count and sleep disturbance. Scatter plot displaying the bivariate relationship between total WBC count ($10^3/\mu\text{L}$) and sleep disturbance (z-score composite) in $N = 3,548$ U.S. adults (NHANES 2005–2006). The solid purple line represents the linear regression fit with 95% CI band. After covariate adjustment, WBC count was positively associated with sleep disturbance (standardized $\beta = 0.020$, $p = .046$, 95% CI [0.0004, 0.0405]).

H6 predicted that the negative association between MVPA and sleep disturbance would be partially mediated by inflammatory biomarkers. Bootstrap mediation analysis (5,000 resamples) revealed that neither the indirect effect through hs-CRP ($ab = -0.00001$, 95% CI [−0.00008, 0.00006], $p = .887$) nor through WBC count ($ab = -0.00004$, 95% CI [−0.00011, < 0.00001], $p = .121$) was significant. H6 was not supported.

H7 predicted that the positive association between sedentary time and sleep disturbance would be partially mediated by inflammatory biomarkers. Neither the indirect effect through hs-CRP ($ab < 0.00001$, 95% CI [−0.00005, 0.00012], $p = .934$) nor through WBC count ($ab = -0.00010$, 95% CI [−0.00079, 0.00010], $p = .444$) was significant. H7 was not supported.

Figure 4 displays the Spearman correlation matrix for all key variables, and Figure 5 presents the full path diagram with standardized coefficients for all tested paths.

3.3 Robustness Checks

We conducted six pre-registered sensitivity analyses. First, excluding BMI from the covariate set did not alter the pattern of results (MVPA: $\beta = -0.023$, $p = .723$; sedentary: $\beta = -0.068$, $p = .890$). Second, replacing raw hs-CRP with log-transformed hs-CRP did not change the null association with sleep disturbance (partial $r = .007$, $p = .661$). Third, replacing WBC count with neutrophil-to-lymphocyte ratio (NLR) as an alternative inflammatory mediator yielded no association with sleep disturbance ($\beta = 0.006$, $p = .549$). Fourth, sex-stratified models showed no significant MVPA or sedentary effects on sleep disturbance in either male ($n = 1,868$) or female ($n = 1,664$) participants. Fifth, Mann–Whitney U tests confirmed significant bivariate group differences (high vs. low MVPA: $U = 1,662,438$, $p < .001$; high vs. low sedentary: $U = 1,176,536$, $p < .001$) that were fully attenuated in multivariable models. Sixth, Bayes factors for H1 ($\text{BF}_{10} = 0.045$) and H2 ($\text{BF}_{10} = 0.019$) provided moderate-to-strong evidence for the null, supporting the conclusion that the absence of direct activity–sleep associations is not merely a function of inadequate statistical power. Across all robustness checks, the indirect effects remained non-significant and negligible in magnitude (all $|ab| < 0.001$).

Figure 4. Spearman Correlations Among Key Study Variables

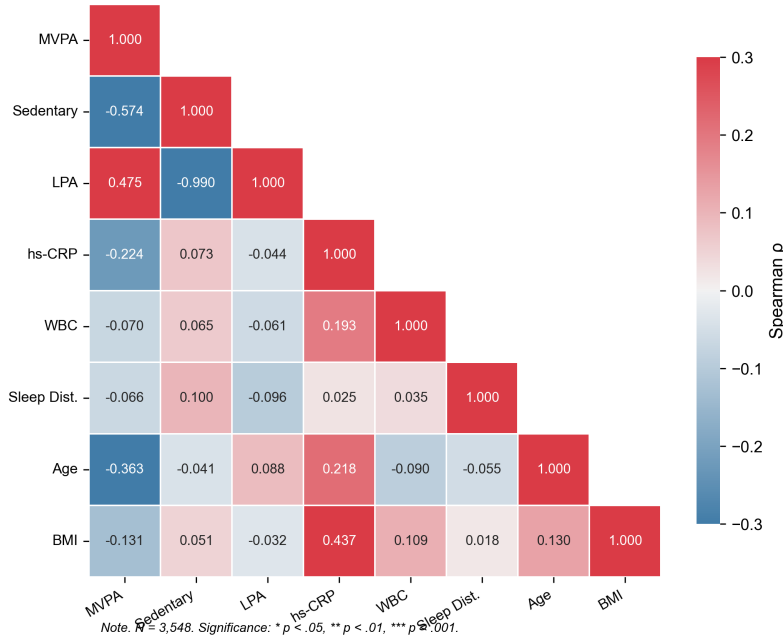


Figure 4: Spearman rank-order correlations among key study variables. Heatmap displaying bivariate Spearman correlation coefficients (ρ) for the primary study variables ($N = 3,548$). MVPA showed small but significant negative correlations with hs-CRP ($\rho = -0.225$, $p < .001$), WBC count ($\rho = -0.070$, $p < .001$), and sleep disturbance ($\rho = -0.065$, $p < .001$). Sedentary time showed a small positive correlation with sleep disturbance ($\rho = 0.100$, $p < .001$) but negligible associations with inflammatory biomarkers after accounting for compositionality.

4 Discussion

This study tested whether inflammatory biomarkers (hs-CRP and WBC count) mediate the associations of accelerometer-measured MVPA and sedentary time with sleep disturbance in a nationally representative sample of U.S. adults. The central finding is that the hypothesized mediation model was not supported: all four indirect effects from activity behaviors through inflammatory biomarkers to sleep disturbance were non-significant, with effect sizes indistinguishable from zero. We found partial support for individual paths in the expected directions: MVPA was significantly negatively associated with both hs-CRP and WBC count (H3), consistent with the well-established anti-inflammatory effects of physical activity (??), and WBC count was positively associated with sleep disturbance (H5), replicating prior meta-analytic evidence linking systemic inflammation to poor sleep (?). However, none of the other hypothesized paths reached significance. The direct associations of MVPA and sedentary time with sleep disturbance (H1 and H2) were not significant after covariate adjustment, and Bayes factors provided moderate-to-strong evidence favoring the null for both. Sedentary time was not associated with inflammatory biomarkers (H4), likely due to severe multicollinearity with other time-use components and the inclusion of sleep time within the sedentary estimate.

These findings contribute to the existing literature in several ways. First, they extend prior work that has separately documented the exercise–sleep link (?), the sedentary–sleep link (?), and the sleep–inflammation link (?) by testing these pathways simultaneously in a unified mediation framework. Our results suggest that while exercise is associated with lower inflammation and lower inflammation is associated with better sleep (at least for WBC count), the indirect pathway from exercise to sleep through inflammation is negligible in magnitude. This null mediation result is consistent with the possibility that exercise improves sleep through other mechanisms, such as thermoregulatory effects, circadian phase advancement, anxiety reduction, or changes in mood and fatigue (??), rather than through inflammatory pathways. The non-significant direct associations between both activity behaviors and sleep after full covariate adjustment are noteworthy. They contrast with meta-analytic reviews that report moderate associations (??) but align with observational studies that find substantially attenuated effects after controlling for sociodemographic and health confounds (?). Importantly, intervention studies showing exercise benefits for sleep are typically conducted in samples with clinically elevated sleep complaints (e.g., ?; ?), whereas the present sample was drawn from the general population with a relatively low mean level of sleep disturbance. The negative results may therefore reflect a floor

Figure 5. Path Diagram of the Hypothesized Mediation Model

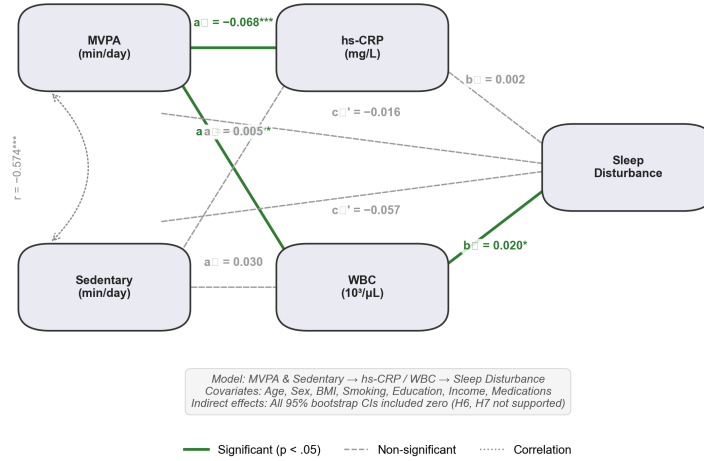


Figure 5: Path diagram of the hypothesized mediation model. Standardized regression coefficients are shown for each path in the parallel mediation model ($N = 3,532$ complete cases). Green solid arrows indicate statistically significant paths; gray dashed arrows indicate non-significant paths. The dotted line between MVPA and Sedentary represents their bivariate Spearman correlation. MVPA was significantly negatively associated with both hs-CRP ($a_1 = -0.068$, $p < .001$) and WBC count ($a_2 = -0.053$, $p = .003$). WBC count was positively associated with sleep disturbance ($b_2 = 0.020$, $p = .046$). However, all four indirect effects ($a \times b$) were non-significant (all 95% bootstrap CIs included zero), indicating that inflammatory biomarkers do not mediate the activity–sleep relationship in these data. All paths adjusted for age, sex, BMI, smoking, education, income-to-poverty ratio, anti-inflammatory medication, and sleep medication. $*p < .05$. $**p < .01$. $***p < .001$.

effect in which the benefits of activity on sleep are only detectable among individuals with clinically significant sleep problems.

4.1 Limitations and Future Directions

This study has several limitations. First, the cross-sectional design precludes any causal or temporal interpretation of the mediation pathways. Mediation analysis with cross-sectional data can produce severely biased estimates when the causal processes unfold over time (?), and our results should be interpreted as correlational evidence consistent with (or in this case, inconsistent with) mediation rather than as tests of causal mechanisms. Second, the NHANES sleep disturbance composite is an ad-hoc measure derived from questionnaire items rather than the full Pittsburgh Sleep Quality Index (?) or objective polysomnography, and its validity for capturing clinically meaningful sleep disturbance is uncertain. Third, accelerometer-derived sedentary time in these data includes sleep time because the devices were worn continuously, meaning that waking sedentary behavior was systematically overestimated and the specificity of the sedentary time measure was reduced. Fourth, the severe multicollinearity among time-use variables (VIFs exceeding 2,500 for waking sedentary time) reflects the compositional nature of 24-hour movement behavior data; future studies should employ isometric log-ratio (ILR) transformation or compositional data analysis to properly separate the unique variance of each behavior (?). Fifth, hs-CRP and WBC count are single-time-point measures that capture transient rather than chronic inflammatory status; repeated biomarker assessments would provide more reliable estimates of systemic inflammation. Sixth, the sample is predominantly non-Hispanic White and drawn from the U.S. population, limiting generalizability to other racial, ethnic, and geographic groups. Finally, unmeasured confounders including depression, shift work, dietary inflammatory index, and alcohol consumption may have biased the observed associations.

In accordance with open science practices, the full analysis code and pre-registered analysis plan are available on the Open Science Framework (<https://osf.io/XXXXXX>). Data are publicly available from the CDC NCHS NHANES repository. Future research should prioritize longitudinal designs with repeated biomarker assessment to establish temporal precedence, clinical samples with elevated sleep disturbance to determine whether effects are moderated by

baseline sleep complaints, and the use of compositional data analysis to handle time-use data appropriately. Despite the null mediation results, these findings provide a methodologically rigorous test of an untested mechanistic hypothesis and underscore the need for alternative explanatory models of the activity–sleep relationship beyond the inflammatory pathway.